

InterpBench

Interpretability Benchmark Report

Accuracy, FGSM Accuracy, PGD Accuracy, Interpretability Score, Interpretability Integral

Model	Dataset	Trainer	Runs
ResNet-18	CIFAR10	PGD	2
ResNet-18	CIFAR10	STANDARD	2
ResNet-18	CUB200	PGD	2
ResNet-18	CUB200	STANDARD	2
ResNet-18	TinyImageNet200	PGD	2
ResNet-18	TinyImageNet200	STANDARD	2
ViT-B/16	CUB200	PGD	2
ViT-B/16	CUB200	STANDARD	2

1 Interpretability Metrics

Figure 1 summarizes the construction of the interpretability metric. Geometric descriptors are aggregated into pairwise and intrinsic geometric burdens, which determine the layer-wise interpretability score and are subsequently integrated over network depth.

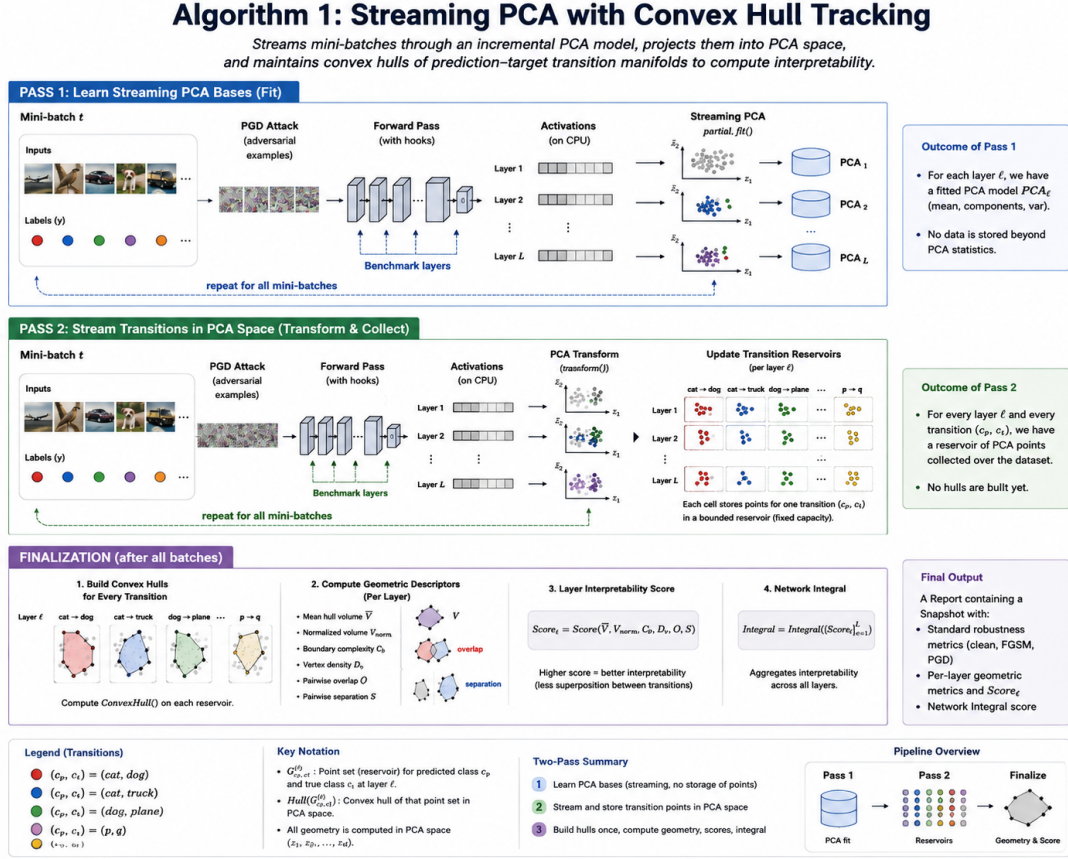


Figure 1: Conceptual construction of the interpretability metric.

Each benchmark layer is represented by geometric descriptors extracted from its valid transition manifolds. Let $\bar{V}$ denote the mean normalized hull volume, I the mean transition-hull intersection, S the mean transition-hull separation, B the mean boundary complexity, and D the mean vertex density. These descriptors capture spatial extent, transition superposition, geometric separation, boundary irregularity, and hull structural complexity.

1.0.1 Interpretability Score

The layer-wise interpretability score is computed from two forms of superposition burden: a pairwise transition-hull burden and an intrinsic geometric burden. The implementation first transforms the mean separation and mean boundary complexity into bounded quantities.

The normalized separation is

$$S_{\text{norm}} = \frac{S}{1 + S}, \quad (1)$$

where S is the mean minimum vertex-to-vertex separation over valid transition-hull pairs. Thus,

$$0 \leq S_{\text{norm}} < 1. \quad (2)$$

The normalized boundary complexity is

$$B_{\text{norm}} = \frac{B}{1+B}, \quad (3)$$

where B is the mean transition-hull boundary complexity.

The pairwise hull burden is defined as

$$H = I(1 + QS_{\text{norm}}), \quad (4)$$

where I is the mean transition-hull overlap. Equivalently,

$$H = I \left(1 + Q \frac{S}{1+S} \right). \quad (5)$$

Thus, overlap increases the burden, while greater geometric separation further amplifies the contribution of overlapping transition hulls.

The multiplicative term

$$A_Q = 1 + QS_{\text{norm}} \quad (6)$$

is referred to as the *separation amplification factor*. Since

$$0 \leq S_{\text{norm}} < 1, \quad (7)$$

its range is

$$1 \leq A_Q < 1 + Q. \quad (8)$$

Thus, Q directly controls the maximum possible amplification of the overlap burden by geometric separation. In particular, larger values of Q make the interpretability score more sensitive to separation among transition-hull pairs with non-zero overlap.

For the present formulation, an effective sensitivity range is

$$10 \leq Q \leq 100, \quad (9)$$

with Q treated as a tunable amplification hyperparameter. This range allows the sensitivity of the score to separation to be systematically evaluated while ensuring that separation provides a substantial signal relative to the baseline overlap contribution.

The intrinsic geometry burden is

$$G = 0.40\bar{V} + 0.30B_{\text{norm}} + 0.30D, \quad (10)$$

where $\bar{V}$ is the mean normalized hull volume and D is the mean vertex density.

For the full evaluation, the superposition burden is the weighted combination

$$\mathcal{B} = \alpha_h H + (1 - \alpha_h)G, \quad 0 \leq \alpha_h \leq 1. \quad (11)$$

Substituting the definitions of H and G gives

$$\mathcal{B} = \alpha_h I \left(1 + Q \frac{S}{1+S} \right) + (1 - \alpha_h) \left(0.40\bar{V} + 0.30 \frac{B}{1+B} + 0.30D \right). \quad (12)$$

The final layer-wise interpretability score is defined as the complement of this superposition burden:

$$\mathcal{I} = 1 - \left[\alpha_h I \left(1 + Q \frac{S}{1+S} \right) + (1 - \alpha_h) \left(0.40\bar{V} + 0.30 \frac{B}{1+B} + 0.30D \right) \right]. \quad (13)$$

Thus, lower overlap and lower intrinsic geometric burden produce a larger interpretability score. For non-zero overlap, greater separation increases the pairwise hull burden through the amplification factor A_Q , with the strength of this effect controlled by Q .

For the light-evaluation mode, pairwise hull geometry is not computed. Instead, the implementation sets the superposition burden directly to the intrinsic geometry burden:

$$\mathcal{B}^{\text{light}} = G. \quad (14)$$

The corresponding interpretability score is therefore

$$S_{\text{interp}}^{\text{light}} = 1 - \left(0.40\bar{V} + 0.30\frac{B}{1+B} + 0.30D \right). \quad (15)$$

Hence, α_h and the pairwise hull quantities I , S , and Q do not affect the score in light-evaluation mode.

1.0.2 Interactive rather than additive

The initial formulation demonstrates stable sensitivity to changes in representational geometry across training. We therefore investigated whether feature superposition should enter the metric additively or instead interact with the intrinsic geometric complexity of the representation.

The initial formulation treats the intrinsic geometry burden G and the pairwise transition-hull burden as additive contributions to the overall superposition burden. Abstracting from the weighting parameter α_h , this corresponds to

$$\mathcal{B}_{\text{add}} = G + O(1 + QS_{\text{norm}}), \quad (16)$$

where O denotes the mean transition-hull overlap and $S_{\text{norm}} = S/(1+S)$.

We investigated an alternative formulation in which transition-hull superposition interacts directly with intrinsic geometric complexity:

$$\mathcal{B}_{\text{int}} = G[1 + O(1 + QS_{\text{norm}})]. \quad (17)$$

The distinction is substantive rather than a rescaling of the output. In the additive formulation, the overlap burden contributes independently of the intrinsic geometry. In the interactive formulation, the effect of transition-hull superposition is modulated by the intrinsic geometric burden G . Thus, the same degree of overlap can produce a different overall burden depending on the geometric complexity of the representation.

This formulation reflects the hypothesis that feature superposition is not independent of intrinsic representation geometry, but emerges from their interaction. The objective is therefore to increase sensitivity and dynamic range by making the scalar score more faithful to the underlying interaction between intrinsic geometric complexity and transition-hull superposition, rather than by simply stretching the numerical output.

The additive and interactive formulations can consequently be compared as alternative models of the same underlying geometric construct:

$$\boxed{\mathcal{B}_{\text{add}} = G + O(1 + QS_{\text{norm}})} \quad \text{versus} \quad \boxed{\mathcal{B}_{\text{int}} = G[1 + O(1 + QS_{\text{norm}})]}. \quad (18)$$

The resulting comparison tests whether the observed fluctuations in the interpretability score can be made more discriminative through a more faithful representation of geometric superposition, while preserving the stability of the metric across training.

1.0.3 Final interactive score formulation

For the full-evaluation mode, the interactive formulation is implemented by using the intrinsic geometric burden G as the base quantity and allowing transition-hull overlap to increase this burden through an interaction with G . The weighting parameter α_h controls the strength of this interaction. Specifically, the interactive hull contribution is defined as

$$H_{\text{int}} = G [1 + O (1 + Q S_{\text{norm}})] , \quad (19)$$

where O is the mean transition-hull overlap and

$$S_{\text{norm}} = \frac{S}{1 + S} . \quad (20)$$

The total superposition burden retains the α_h weighting between the interaction-sensitive contribution and the intrinsic geometric contribution:

$$\mathcal{B} = \alpha_h H_{\text{int}} + (1 - \alpha_h) G . \quad (21)$$

Substituting H_{int} and simplifying gives the equivalent form

$$\boxed{\mathcal{B} = G [1 + \alpha_h O (1 + Q S_{\text{norm}})]} . \quad (22)$$

The final interpretability score is then defined using a reciprocal transformation of the superposition burden:

$$\boxed{\mathcal{I} = \frac{1}{1 + G [1 + \alpha_h O (1 + Q S_{\text{norm}})]}} . \quad (23)$$

Thus, higher intrinsic geometric burden, greater transition-hull overlap, and greater separation among overlapping transition hulls all increase the superposition burden and consequently decrease the interpretability score. The parameter Q controls the amplification of overlap by separation, while α_h controls the strength with which the resulting overlap–separation interaction contributes to the total burden.

For the current implementation,

$$Q = 4005 . \quad (24)$$

This formulation is the operational definition used by the implementation. It should be distinguished from the earlier additive formulation $G + O(1 + Q S_{\text{norm}})$: the present metric explicitly models transition-hull superposition as an interaction with intrinsic geometric complexity rather than as an independent additive burden.

1.1 Interpretability Integral

Let the normalized depths of the L benchmark layers be

$$d_1, d_2, \dots, d_L , \quad (25)$$

with corresponding interpretability scores

$$s_1, s_2, \dots, s_L . \quad (26)$$

The overall interpretability is computed using the trapezoidal rule:

$$\mathcal{I} = \text{trapz}(d_1, \dots, d_L; s_1, \dots, s_L). \quad (27)$$

Equivalently,

$$\mathcal{I} = \sum_{\ell=1}^{L-1} \frac{s_{\ell} + s_{\ell+1}}{2} (d_{\ell+1} - d_{\ell}). \quad (28)$$

For a single benchmark layer,

$$\mathcal{I} = s_1. \quad (29)$$

The resulting quantity is therefore the area under the layer-wise interpretability curve across normalized network depth, rather than the arithmetic mean of the layer scores. The weighting parameter α_h is applied consistently to full-mode layer scores and is inactive in light-evaluation mode.

Table 1: Default training and benchmark hyperparameter configuration.

Parameter	Value	Description
Training Configuration		
Initial learning rate	0.1	Initial optimizer learning rate
Momentum	0.9	Momentum coefficient for SGD updates
Weight decay	5×10^{-4}	L2 regularization coefficient
Scheduler milestones	[100, 150]	Epochs at which the learning rate is decayed
Scheduler decay factor (γ)	0.1	Multiplicative decay applied at each milestone
Batch size	32	Number of training samples per optimization step
Training epochs	100	Total number of training epochs
Benchmark frequency	Every 10 epochs	Frequency of benchmark evaluation during training
PGD Configuration		
Perturbation budget (ϵ)	8/255	Maximum ℓ_∞ perturbation magnitude
Step size (α)	2/255	Gradient ascent step size for adversarial example generation
Iterations	10	Number of PGD optimization steps
Feature Processing and Geometry		
PCA components (k)	10	Number of retained principal components
Maximum reservoir size (M)	128	Maximum number of stored samples per geometric reservoir
Scoring Configuration		
Hull burden amplification factor (Q)	4005	Amplification factor for hull burden
Interpretability balance (α_h)	0.75	Interpolation between overlap-centric and geometry-centric scoring

2 Hyperparameters

The default training and benchmark configurations used throughout the experiments are summarized in Table 1. Unless otherwise stated, these settings are kept fixed across all experiments.

3 Tables

3.1 Summary

Model	Dataset	Trainer	Epoch	Accuracy	PGD	F1-Robust	Score	Integral
ResNet-18	CIFAR10	PGD	100	20.4300 ± 1.7789	50.5250 ± 51.3966	29.0500 ± 6.7495	0.6160 ± 0.1841	0.6112 ± 0.2469
ResNet-18	CIFAR10	STANDARD	100	78.2100 ± 29.6055	14.5150 ± 27.5089	24.3822 ± 37.6880	0.6076 ± 0.0289	0.6086 ± 0.0780
ResNet-18	CUB200	PGD	100	0.6817 ± 1.8640	9.0784 ± 5.7018	1.2660 ± 3.2829	$0.6255 \pm \text{N/A}$	$0.6255 \pm \text{N/A}$
ResNet-18	CUB200	STANDARD	100	29.7031 ± 105.0444	0.0777 ± 0.3289	0.1548 ± 0.6525	0.6259 ± 0.0038	0.6259 ± 0.0047
ResNet-18	TinyImageNet200	PGD	100	0.6550 ± 0.0635	21.0350 ± 5.7813	1.2704 ± 0.1090	0.6264 ± 0.0031	0.6264 ± 0.0033
ResNet-18	TinyImageNet200	STANDARD	100	35.8150 ± 39.0716	2.4450 ± 8.1955	4.5688 ± 14.6930	0.6266 ± 0.0001	0.6267 ± 0.0003
ViT-B/16	CUB200	PGD	100	3.0721 ± 11.1843	7.5078 ± 13.8159	4.1929 ± 8.8926	N/A	N/A
ViT-B/16	CUB200	STANDARD	100	15.7663 ± 52.9608	0.1467 ± 0.5482	0.2907 ± 1.0854	0.6267 ± 0.0007	0.6265 ± 0.0011

3.2 Benchmark Comparison (Best Score)

Model	Dataset	Trainer	Run	Best Epoch	Last Epoch	Acc Best	Acc Last	PGD Best	PGD Last	S Best	S Last	Integral Best	Integral Last
ResNet-18	CIFAR10	PGD	AVE			8.9100 $\pm$ 6.4802	20.4300 $\pm$ 1.7789	9.8900 $\pm$ 0.7624	50.5250 $\pm$ 51.3966	0.6327 $\pm$ 0.0126	0.6160 $\pm$ 0.1841	0.6325 $\pm$ 0.0117	0.6112 $\pm$ 0.2469
			0	0	100	9.4200	20.2900	9.8300	54.5700	0.6317	0.6305	0.6316	0.6306
			1	0	100	8.4000	20.5700	9.9500	46.4800	0.6337	0.6015	0.6334	0.5917
ResNet-18	CIFAR10	STANDARD	AVE			10.3200 $\pm$ 10.1650	78.2100 $\pm$ 29.6055	9.0950 $\pm$ 11.3721	14.5150 $\pm$ 27.5089	0.6329 $\pm$ 0.0126	0.6076 $\pm$ 0.0289	0.6329 $\pm$ 0.0129	0.6086 $\pm$ 0.0780
			0	0	100	11.1200	80.5400	8.2000	12.3500	0.6320	0.6053	0.6319	0.6147
			1	0	100	9.5200	75.8800	9.9900	16.6800	0.6339	0.6099	0.6339	0.6024
ResNet-18	CUB200	PGD	AVE			0.5868 $\pm$ 0.2193	0.6817 $\pm$ 1.8640	7.4128 $\pm$ 15.2413	9.0784 $\pm$ 5.7018	0.6274 $\pm$ 0.0062	0.6255 $\pm$ N/A	0.6274 $\pm$ 0.0062	0.6255 $\pm$ N/A
			0	40	100	0.6041	0.5350	6.2133	8.6296	0.6279	nan	0.6279	nan
			1	70	100	0.5696	0.8284	8.6124	9.5271	0.6269	0.6255	0.6269	0.6255
ResNet-18	CUB200	STANDARD	AVE			19.1664 $\pm$ 238.9266	29.7031 $\pm$ 105.0444	0.1294 $\pm$ 0.9868	0.0777 $\pm$ 0.3289	0.6266 $\pm$ 0.0046	0.6259 $\pm$ 0.0038	0.6266 $\pm$ 0.0043	0.6259 $\pm$ 0.0047
			0	0	100	0.3624	21.4360	0.2071	0.1036	0.6270	0.6256	0.6270	0.6256
			1	100	100	37.9703	37.9703	0.0518	0.0518	0.6262	0.6262	0.6263	0.6263
ResNet-18	TinyImageNet200	PGD	AVE			0.5400 $\pm$ 0.2541	0.6550 $\pm$ 0.0635	11.1050 $\pm$ 134.6222	21.0350 $\pm$ 5.7813	0.6282 $\pm$ 0.0007	0.6264 $\pm$ 0.0031	0.6285 $\pm$ 0.0040	0.6264 $\pm$ 0.0033
			0	0	100	0.5600	0.6600	0.5100	20.5800	0.6282	0.6262	0.6288	0.6262
			1	30	100	0.5200	0.6500	21.7000	21.4900	0.6281	0.6266	0.6281	0.6267
ResNet-18	TinyImageNet200	STANDARD	AVE			39.5900 $\pm$ 6.8614	35.8150 $\pm$ 39.0716	2.1750 $\pm$ 4.2566	2.4450 $\pm$ 8.1955	0.6277 $\pm$ 0.0082	0.6266 $\pm$ 0.0001	0.6275 $\pm$ 0.0100	0.6267 $\pm$ 0.0003
			0	80	100	39.0500	38.8900	2.5100	3.0900	0.6270	0.6266	0.6267	0.6267
			1	70	100	40.1300	32.7400	1.8400	1.8000	0.6283	0.6266	0.6283	0.6267
ViT-B/16	CUB200	PGD	AVE			3.1671 $\pm$ 0.1096	3.0721 $\pm$ 11.1843	5.3245 $\pm$ 10.8553	7.5078 $\pm$ 13.8159	0.6279 $\pm$ 0.0053		0.6277 $\pm$ 0.0034	
			0	90	100	3.1757	3.9524	4.4701	6.4204	0.6283	nan	0.6280	nan
			1	40	100	3.1584	2.1919	6.1788	8.5951	0.6274	nan	0.6275	nan
ViT-B/16	CUB200	STANDARD	AVE			5.4021 $\pm$ 60.3073	15.7663 $\pm$ 52.9608	0.2848 $\pm$ 2.9605	0.1467 $\pm$ 0.5482	0.6279 $\pm$ 0.0039	0.6267 $\pm$ 0.0007	0.6276 $\pm$ 0.0033	0.6265 $\pm$ 0.0011
			0	10	100	0.6559	19.9344	0.5178	0.1899	0.6282	0.6267	0.6278	0.6264
			1	80	100	10.1484	11.5982	0.0518	0.1036	0.6276	0.6268	0.6273	0.6266

3.3 Benchmark Comparison (Best Integral)

Model	Dataset	Trainer	Run	Best Epoch	Last Epoch	Acc Best	Acc Last	PGD Best	PGD Last	Score Best	Score Last	Integral Best	Integral Last
ResNet-18	CIFAR10	PGD	AVE			8.9100 $\pm$ 6.4802	20.4300 $\pm$ 1.7789	9.8900 $\pm$ 0.7624	50.5250 $\pm$ 51.3966	0.6327 $\pm$ 0.0126	0.6160 $\pm$ 0.1841	0.6325 $\pm$ 0.0117	0.6112 $\pm$ 0.2469
			0	0	100	9.4200	20.2900	9.8300	54.5700	0.6317	0.6305	0.6316	0.6306
			1	0	100	8.4000	20.5700	9.9500	46.4800	0.6337	0.6015	0.6334	0.5917
ResNet-18	CIFAR10	STANDARD	AVE			41.6950 $\pm$ 408.8221	78.2100 $\pm$ 29.6055	12.3300 $\pm$ 29.7325	14.5150 $\pm$ 27.5089	0.6328 $\pm$ 0.0145	0.6076 $\pm$ 0.0289	0.6330 $\pm$ 0.0122	0.6086 $\pm$ 0.0780
			0	40	100	73.8700	80.5400	14.6700	12.3500	0.6316	0.6053	0.6320	0.6147
			1	0	100	9.5200	75.8800	9.9900	16.6800	0.6339	0.6099	0.6339	0.6024
ResNet-18	CUB200	PGD	AVE			0.5868 $\pm$ 0.2193	0.6817 $\pm$ 1.8640	7.4128 $\pm$ 15.2413	9.0784 $\pm$ 5.7018	0.6274 $\pm$ 0.0062	0.6255 $\pm$ N/A	0.6274 $\pm$ 0.0062	0.6255 $\pm$ N/A
			0	40	100	0.6041	0.5350	6.2133	8.6296	0.6279	nan	0.6279	nan
			1	70	100	0.5696	0.8284	8.6124	9.5271	0.6269	0.6255	0.6269	0.6255
ResNet-18	CUB200	STANDARD	AVE			19.1664 $\pm$ 238.9266	29.7031 $\pm$ 105.0444	0.1294 $\pm$ 0.9868	0.0777 $\pm$ 0.3289	0.6266 $\pm$ 0.0046	0.6259 $\pm$ 0.0038	0.6266 $\pm$ 0.0043	0.6259 $\pm$ 0.0047
			0	0	100	0.3624	21.4360	0.2071	0.1036	0.6270	0.6256	0.6270	0.6256
			1	100	100	37.9703	37.9703	0.0518	0.0518	0.6262	0.6262	0.6263	0.6263
ResNet-18	TinyImageNet200	PGD	AVE			0.5400 $\pm$ 0.2541	0.6550 $\pm$ 0.0635	11.1050 $\pm$ 134.6222	21.0350 $\pm$ 5.7813	0.6282 $\pm$ 0.0007	0.6264 $\pm$ 0.0031	0.6285 $\pm$ 0.0040	0.6264 $\pm$ 0.0033
			0	0	100	0.5600	0.6600	0.5100	20.5800	0.6282	0.6262	0.6288	0.6262
			1	30	100	0.5200	0.6500	21.7000	21.4900	0.6281	0.6266	0.6281	0.6267
ResNet-18	TinyImageNet200	STANDARD	AVE			39.1450 $\pm$ 12.5156	35.8150 $\pm$ 39.0716	2.4350 $\pm$ 7.5602	2.4450 $\pm$ 8.1955	0.6275 $\pm$ 0.0098	0.6266 $\pm$ 0.0001	0.6277 $\pm$ 0.0084	0.6267 $\pm$ 0.0003
			0	10	100	38.1600	38.8900	3.0300	3.0900	0.6268	0.6266	0.6270	0.6267
			1	70	100	40.1300	32.7400	1.8400	1.8000	0.6283	0.6266	0.6283	0.6267
ViT-B/16	CUB200	PGD	AVE			3.1671 $\pm$ 0.1096	3.0721 $\pm$ 11.1843	5.3245 $\pm$ 10.8553	7.5078 $\pm$ 13.8159	0.6279 $\pm$ 0.0053		0.6277 $\pm$ 0.0034	
			0	90	100	3.1757	3.9524	4.4701	6.4204	0.6283	nan	0.6280	nan
			1	40	100	3.1584	2.1919	6.1788	8.5951	0.6274	nan	0.6275	nan
ViT-B/16	CUB200	STANDARD	AVE			5.4021 $\pm$ 60.3073	15.7663 $\pm$ 52.9608	0.2848 $\pm$ 2.9605	0.1467 $\pm$ 0.5482	0.6279 $\pm$ 0.0039	0.6267 $\pm$ 0.0007	0.6276 $\pm$ 0.0033	0.6265 $\pm$ 0.0011
			0	10	100	0.6559	19.9344	0.5178	0.1899	0.6282	0.6267	0.6278	0.6264
			1	80	100	10.1484	11.5982	0.0518	0.1036	0.6276	0.6268	0.6273	0.6266

3.4 Overall Statistics

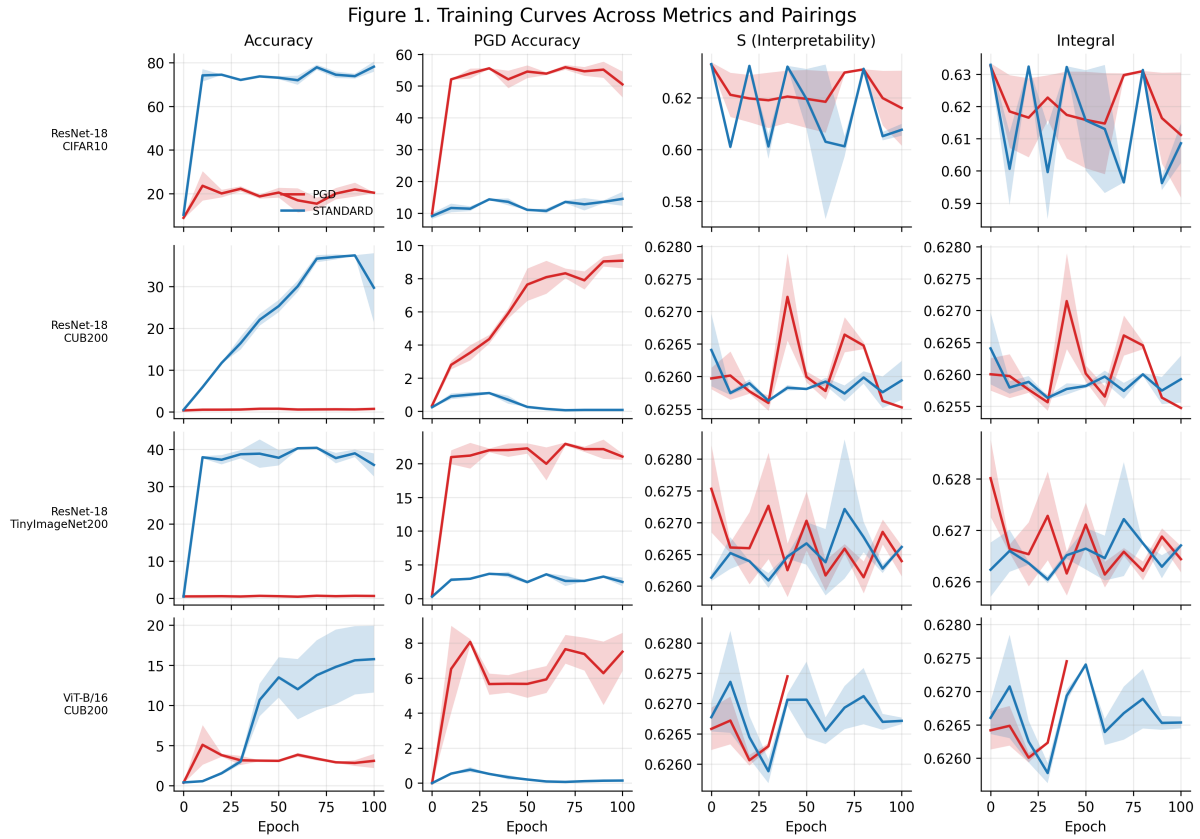
Model	Dataset	Trainer	Epoch	Level	Accuracy	PGD	F1-Robust	Score	Integral
ResNet-18	CIFAR10	PGD	0	Overall	8.9100e+00 $\pm$ 6.4802e+00	9.8900e+00 $\pm$ 7.6237e-01	9.3651e+00 $\pm$ 3.2471e+00	6.3269e-01 $\pm$ 1.2570e-02	6.3251e-01 $\pm$ 1.1652e-02
				Layer 0				6.3370e-01 $\pm$ 1.6095e-02	6.3251e-01 $\pm$ 1.1652e-02
				Layer 1				6.3195e-01 $\pm$ 1.3139e-02	6.3251e-01 $\pm$ 1.1652e-02
				Layer 2				6.3238e-01 $\pm$ 6.4902e-03	6.3251e-01 $\pm$ 1.1652e-02
				Layer 3				6.3273e-01 $\pm$ 1.4558e-02	6.3251e-01 $\pm$ 1.1652e-02
ResNet-18	CIFAR10	PGD	10	Overall	2.3610e+01 $\pm$ 8.5386e+01	5.2100e+01 $\pm$ 1.1436e+00	3.1932e+01 $\pm$ 8.1735e+01	6.2111e-01 $\pm$ 1.0838e-01	6.1845e-01 $\pm$ 1.4342e-01
				Layer 0				6.2761e-01 $\pm$ 1.4690e-02	6.1845e-01 $\pm$ 1.4342e-01
				Layer 1				5.9560e-01 $\pm$ 4.3755e-01	6.1845e-01 $\pm$ 1.4342e-01
				Layer 2				6.3066e-01 $\pm$ 1.0558e-02	6.1845e-01 $\pm$ 1.4342e-01
				Layer 3				6.3059e-01 $\pm$ 8.1704e-03	6.1845e-01 $\pm$ 1.4342e-01
ResNet-18	CIFAR10	PGD	20	Overall	2.0095e+01 $\pm$ 2.2045e+01	5.3950e+01 $\pm$ 1.9822e+01	2.9206e+01 $\pm$ 2.0489e+01	6.1970e-01 $\pm$ 1.1588e-01	6.1653e-01 $\pm$ 1.5565e-01
				Layer 0				6.2893e-01 $\pm$ 2.3617e-03	6.1653e-01 $\pm$ 1.5565e-01
				Layer 1				5.9061e-01 $\pm$ 4.7572e-01	6.1653e-01 $\pm$ 1.5565e-01
				Layer 2				6.2975e-01 $\pm$ 5.3355e-03	6.1653e-01 $\pm$ 1.5565e-01
				Layer 3				6.2952e-01 $\pm$ 4.5134e-03	6.1653e-01 $\pm$ 1.5565e-01
ResNet-18	CIFAR10	PGD	30	Overall	2.2230e+01 $\pm$ 1.5247e+01	5.5570e+01 $\pm$ 1.2706e+00	3.1736e+01 $\pm$ 1.5354e+01	6.1900e-01 $\pm$ 1.3566e-01	6.2275e-01 $\pm$ 8.8321e-02
				Layer 0				6.2956e-01 $\pm$ 8.3619e-03	6.2275e-01 $\pm$ 8.8321e-02
				Layer 1				6.3007e-01 $\pm$ 4.7953e-03	6.2275e-01 $\pm$ 8.8321e-02
				Layer 2				6.3042e-01 $\pm$ 7.9118e-03	6.2275e-01 $\pm$ 8.8321e-02
				Layer 3				5.8595e-01 $\pm$ 5.6370e-01	6.2275e-01 $\pm$ 8.8321e-02
ResNet-18	CIFAR10	PGD	40	Overall	1.8750e+01 $\pm$ 1.2325e+01	5.2110e+01 $\pm$ 3.4815e+01	2.7519e+01 $\pm$ 8.4370e+00	6.2043e-01 $\pm$ 1.2795e-01	6.1739e-01 $\pm$ 1.7142e-01
				Layer 0				6.2841e-01 $\pm$ 1.2018e-02	6.1739e-01 $\pm$ 1.7142e-01
				Layer 1				5.9157e-01 $\pm$ 5.0716e-01	6.1739e-01 $\pm$ 1.7142e-01
				Layer 2				6.3103e-01 $\pm$ 9.5709e-03	6.1739e-01 $\pm$ 1.7142e-01
				Layer 3				6.3070e-01 $\pm$ 7.0700e-03	6.1739e-01 $\pm$ 1.7142e-01
ResNet-18	CIFAR10	PGD	50	Overall	2.0495e+01 $\pm$ 2.8652e+01	5.4545e+01 $\pm$ 2.5222e+01	2.9668e+01 $\pm$ 2.6520e+01	6.1954e-01 $\pm$ 1.4212e-01	6.1583e-01 $\pm$ 1.9023e-01
				Layer 0				6.3032e-01 $\pm$ 4.0877e-03	6.1583e-01 $\pm$ 1.9023e-01
				Layer 1				6.3052e-01 $\pm$ 3.9091e-03	6.1583e-01 $\pm$ 1.9023e-01
				Layer 2				5.8628e-01 $\pm$ 5.6900e-01	6.1583e-01 $\pm$ 1.9023e-01
				Layer 3				6.3105e-01 $\pm$ 3.6412e-04	6.1583e-01 $\pm$ 1.9023e-01
ResNet-18	CIFAR10	PGD	60	Overall	1.6930e+01 $\pm$ 7.0519e+01	5.3905e+01 $\pm$ 5.4001e+00	2.5284e+01 $\pm$ 8.2812e+01	6.1841e-01 $\pm$ 1.5103e-01	6.1471e-01 $\pm$ 1.9922e-01
				Layer 0				6.2970e-01 $\pm$ 3.9057e-03	6.1471e-01 $\pm$ 1.9922e-01
				Layer 1				6.3022e-01 $\pm$ 4.9410e-03	6.1471e-01 $\pm$ 1.9922e-01
				Layer 2				5.8439e-01 $\pm$ 5.8624e-01	6.1471e-01 $\pm$ 1.9922e-01
				Layer 3				6.2932e-01 $\pm$ 9.0520e-03	6.1471e-01 $\pm$ 1.9922e-01
ResNet-18	CIFAR10	PGD	70	Overall	1.5400e+01 $\pm$ 3.5577e+01	5.5905e+01 $\pm$ 9.4661e+00	2.4032e+01 $\pm$ 4.4695e+01	6.2970e-01 $\pm$ 7.1020e-03	6.2973e-01 $\pm$ 7.4791e-03
				Layer 0				6.2940e-01 $\pm$ 7.6011e-03	6.2973e-01 $\pm$ 7.4791e-03
				Layer 1				6.2927e-01 $\pm$ 1.1962e-02	6.2973e-01 $\pm$ 7.4791e-03
				Layer 2				6.3034e-01 $\pm$ 4.5049e-03	6.2973e-01 $\pm$ 7.4791e-03
				Layer 3				6.2979e-01 $\pm$ 4.3403e-03	6.2973e-01 $\pm$ 7.4791e-03
ResNet-18	CIFAR10	PGD	80	Overall	2.0055e+01 $\pm$ 3.2846e+01	5.4635e+01 $\pm$ 1.7598e+01	2.9278e+01 $\pm$ 3.7729e+01	6.3092e-01 $\pm$ 1.3674e-03	6.3096e-01 $\pm$ 1.2075e-03
				Layer 0				6.3056e-01 $\pm$ 2.4102e-03	6.3096e-01 $\pm$ 1.2075e-03
				Layer 1				6.3094e-01 $\pm$ 1.0385e-03	6.3096e-01 $\pm$ 1.2075e-03
				Layer 2				6.3114e-01 $\pm$ 7.3685e-04	6.3096e-01 $\pm$ 1.2075e-03
				Layer 3				6.3102e-01 $\pm$ 1.2839e-03	6.3096e-01 $\pm$ 1.2075e-03
ResNet-18	CIFAR10	PGD	90	Overall	2.1915e+01 $\pm$ 4.0088e+01	5.5155e+01 $\pm$ 3.2210e+01	3.1305e+01 $\pm$ 4.6329e+01	6.1985e-01 $\pm$ 1.3394e-01	6.1639e-01 $\pm$ 1.7920e-01
				Layer 0				6.2997e-01 $\pm$ 5.7503e-03	6.1639e-01 $\pm$ 1.7920e-01
				Layer 1				6.2987e-01 $\pm$ 9.3558e-03	6.1639e-01 $\pm$ 1.7920e-01
				Layer 2				5.8909e-01 $\pm$ 5.3013e-01	6.1639e-01 $\pm$ 1.7920e-01
				Layer 3				6.3046e-01 $\pm$ 2.0063e-03	6.1639e-01 $\pm$ 1.7920e-01
ResNet-18	CIFAR10	PGD	100	Overall	2.0430e+01 $\pm$ 1.7789e+00	5.0525e+01 $\pm$ 5.1397e+01	2.9050e+01 $\pm$ 6.7495e+00	6.1598e-01 $\pm$ 1.8407e-01	6.1116e-01 $\pm$ 2.4689e-01
				Layer 0				6.3006e-01 $\pm$ 5.0990e-03	6.1116e-01 $\pm$ 2.4689e-01
				Layer 1				6.3053e-01 $\pm$ 5.2344e-04	6.1116e-01 $\pm$ 2.4689e-01

3.5 Significance Tests (whether PGD trainer changes metric relative to STANDARD trainer)

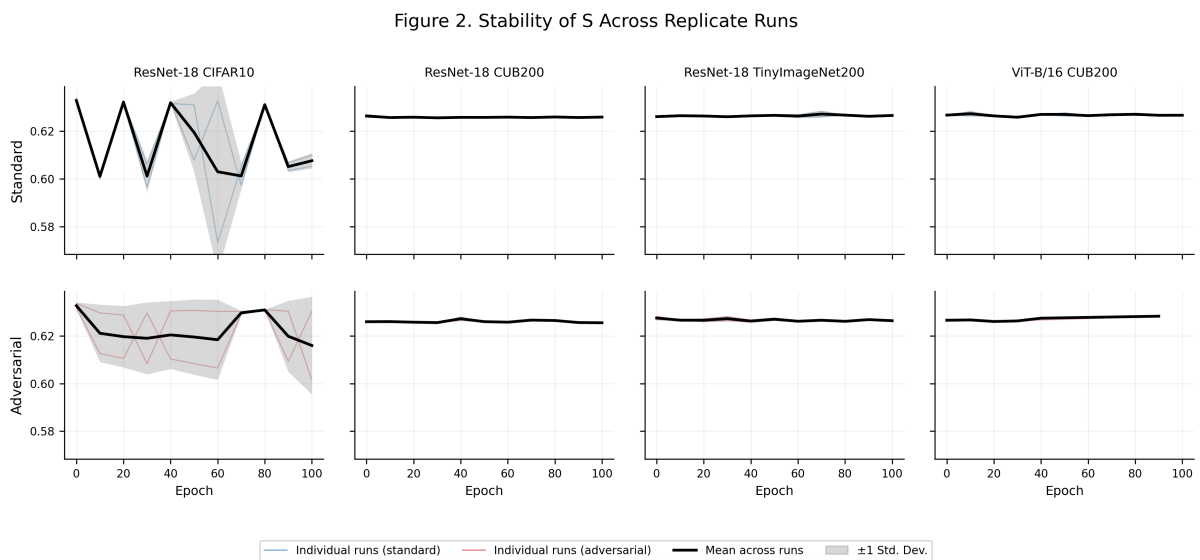
Pairing	Metric	Test	Test statistic	p -value	Significant ($\alpha = 0.05$)
ResNet-18/CIFAR10	Accuracy	Wilcoxon	0.0000	0.5	No
	PGD Accuracy	Wilcoxon	0.0000	0.5	No
	S	Wilcoxon	1.0000	1	No
	Integral	Wilcoxon	1.0000	1	No
ResNet-18/CUB200	Accuracy	Wilcoxon	0.0000	0.5	No
	PGD Accuracy	Wilcoxon	0.0000	0.5	No
	S		nan	nan	No
	Integral		nan	nan	No
ResNet-18/TinyImageNet200	Accuracy	Wilcoxon	0.0000	0.5	No
	PGD Accuracy	Wilcoxon	0.0000	0.5	No
	S	Wilcoxon	1.0000	1	No
	Integral	Wilcoxon	1.0000	1	No
ViT-B/16/CUB200	Accuracy	Wilcoxon	0.0000	0.5	No
	PGD Accuracy	Wilcoxon	0.0000	0.5	No
	S		nan	nan	No
	Integral		nan	nan	No

4 Figures

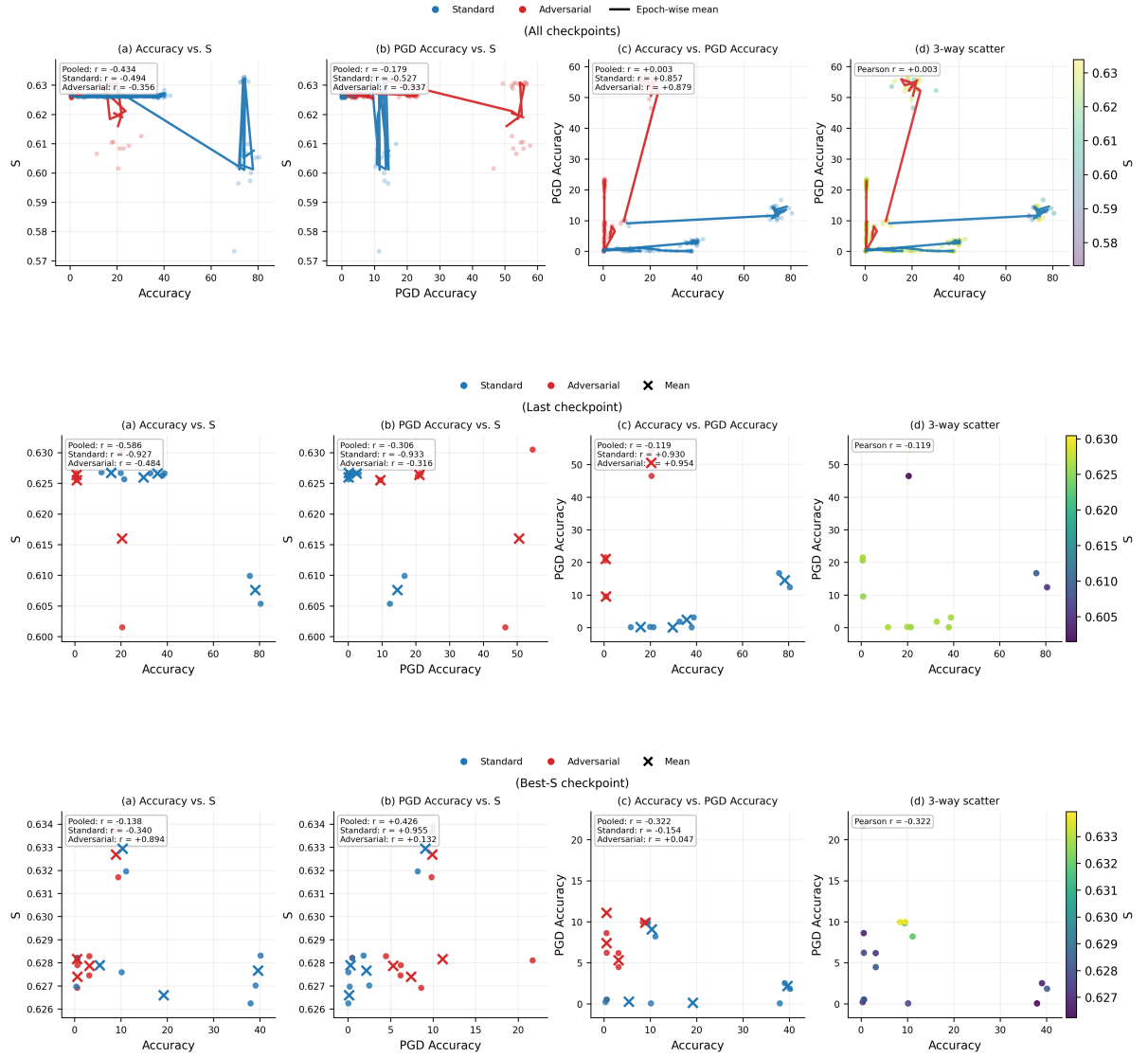
4.1 Training Curves



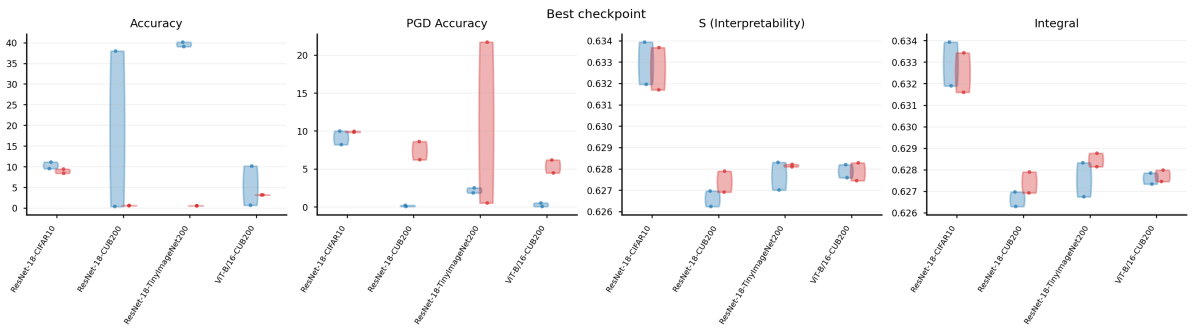
4.2 Performance Stability

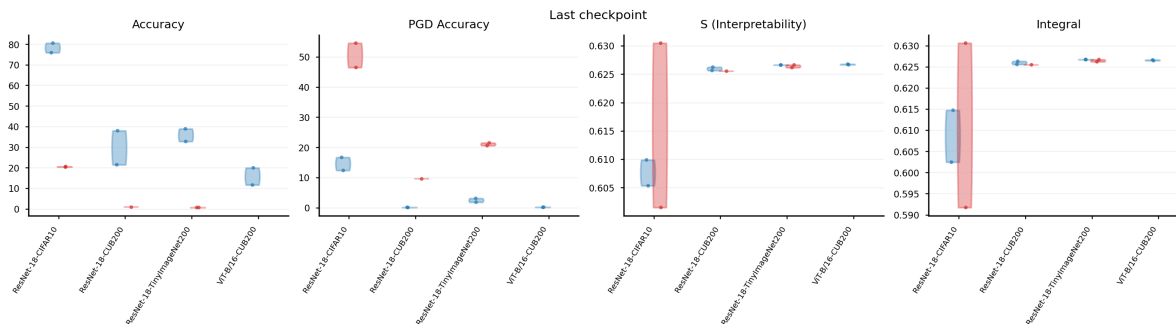


4.3 Correlation / Trade-off Analysis

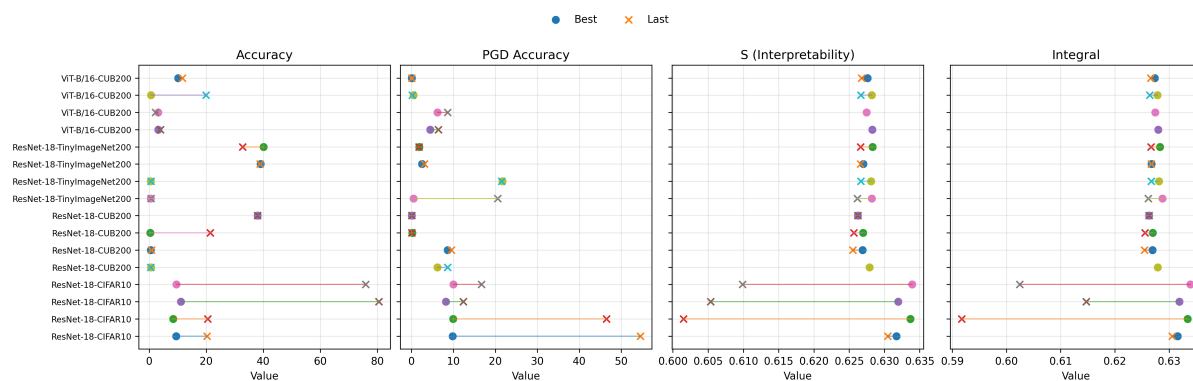


4.4 Distribution Analysis

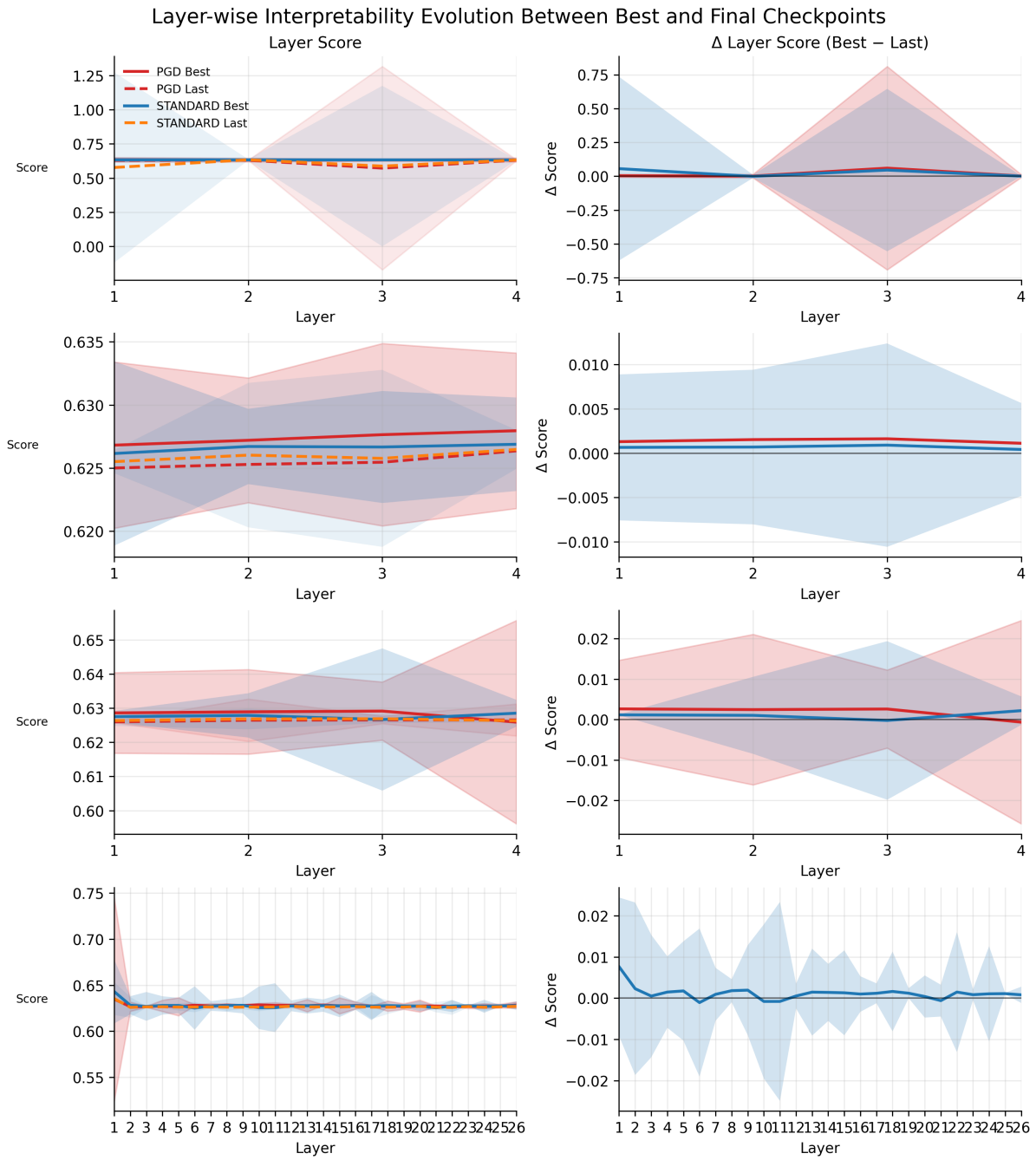




4.5 Best vs Last Interpretability



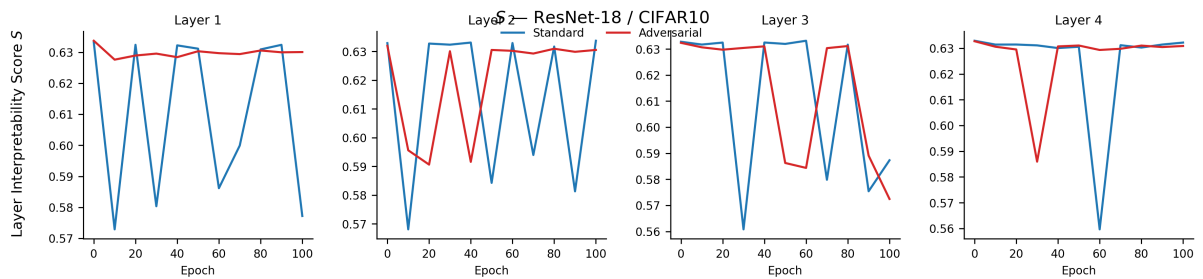
4.6 Best vs Last Checkpoints

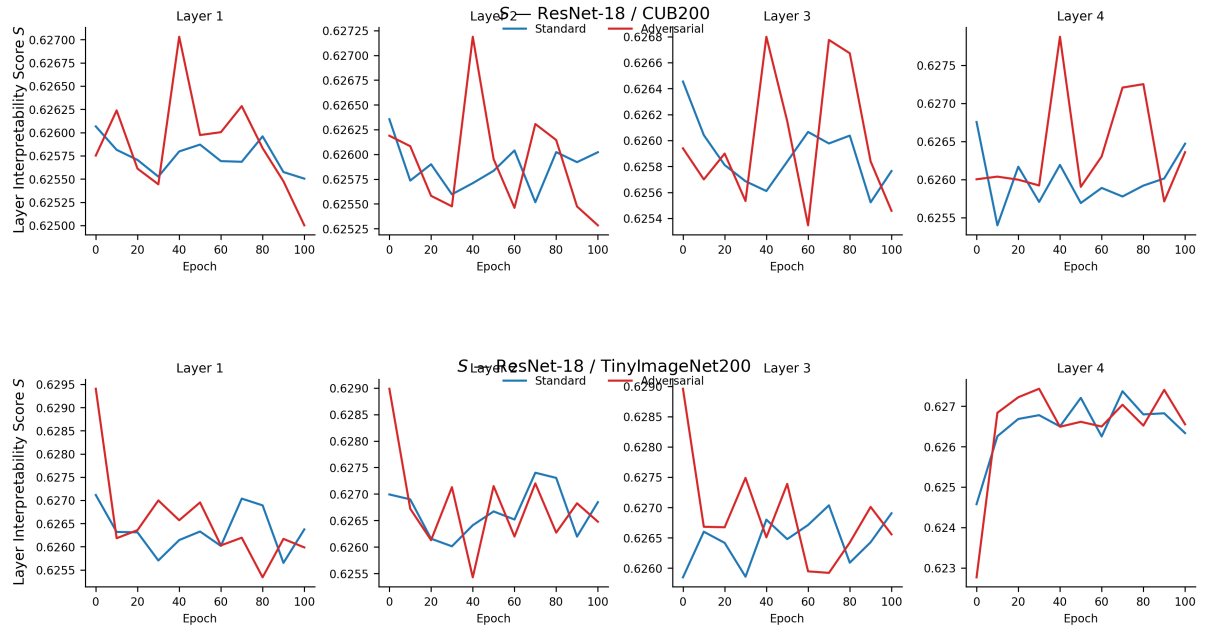


4.7 Layer-wise Scores



4.8 Layer-wise Training Curves

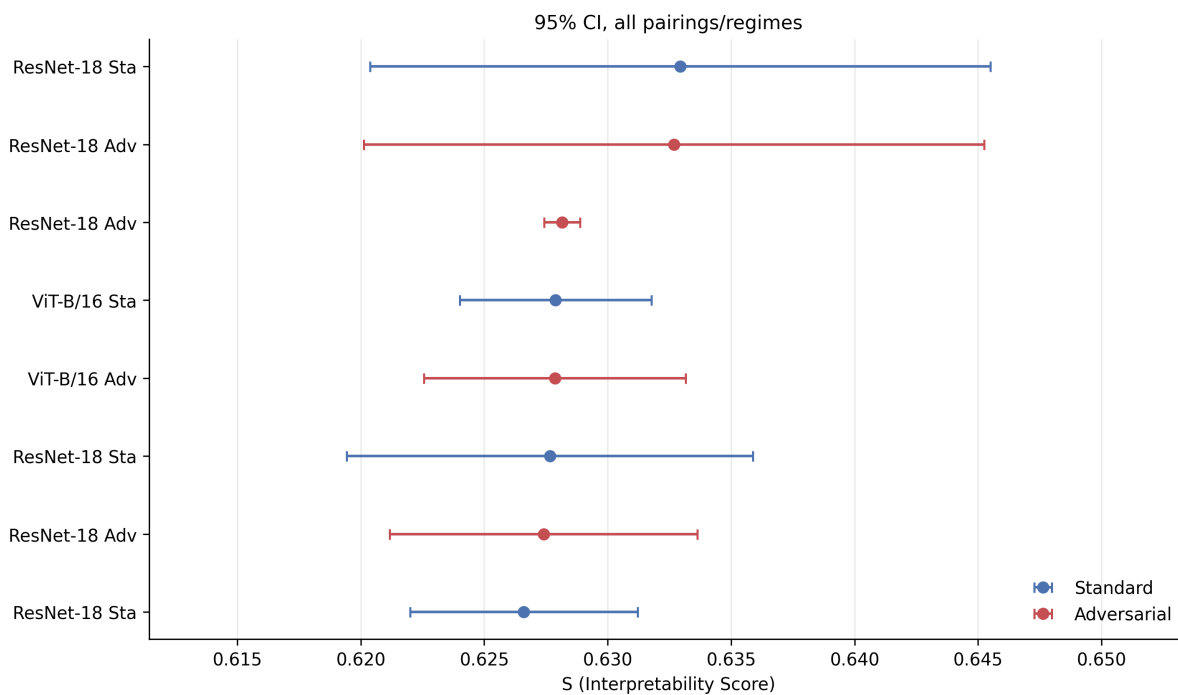




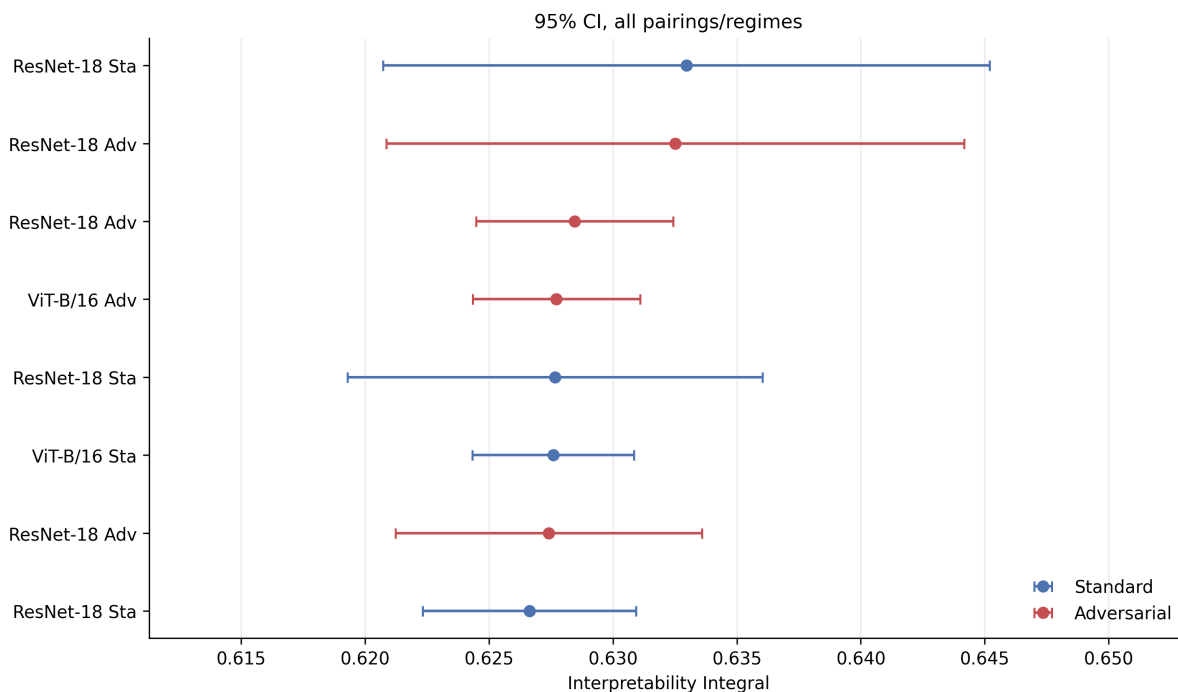
S — ViT-B/16 / CUB200



4.9 Leaderboard S

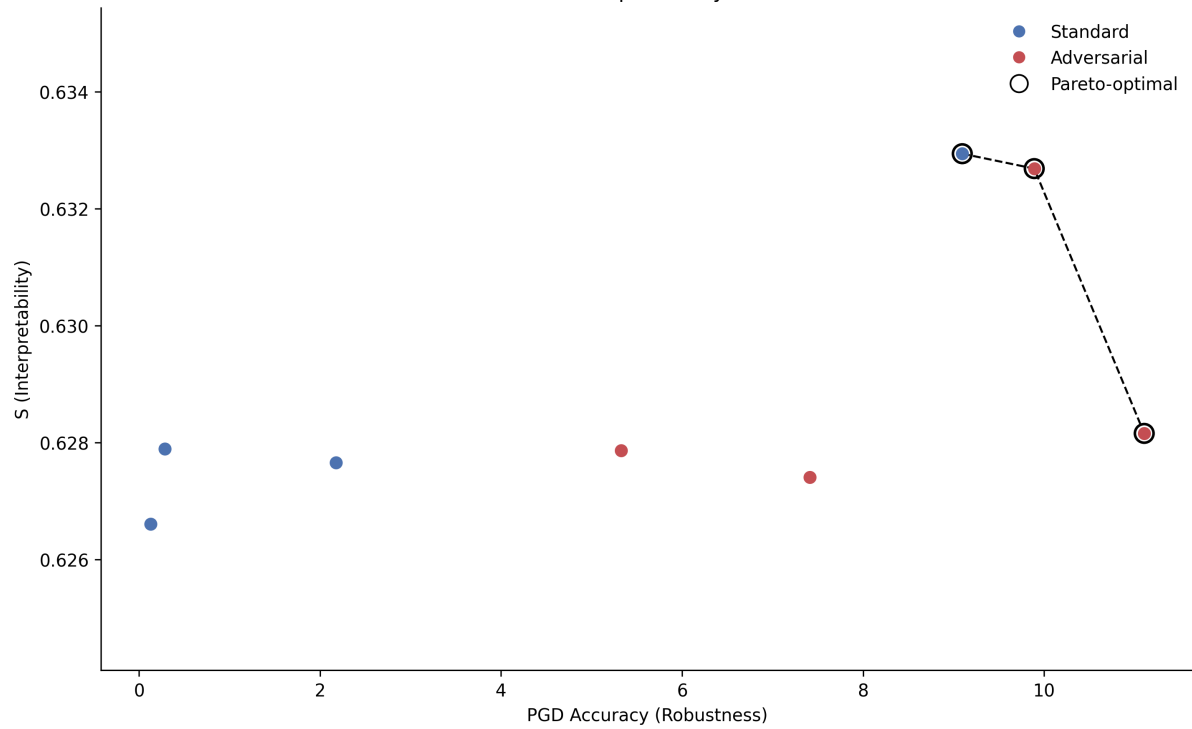


4.10 Leaderboard I

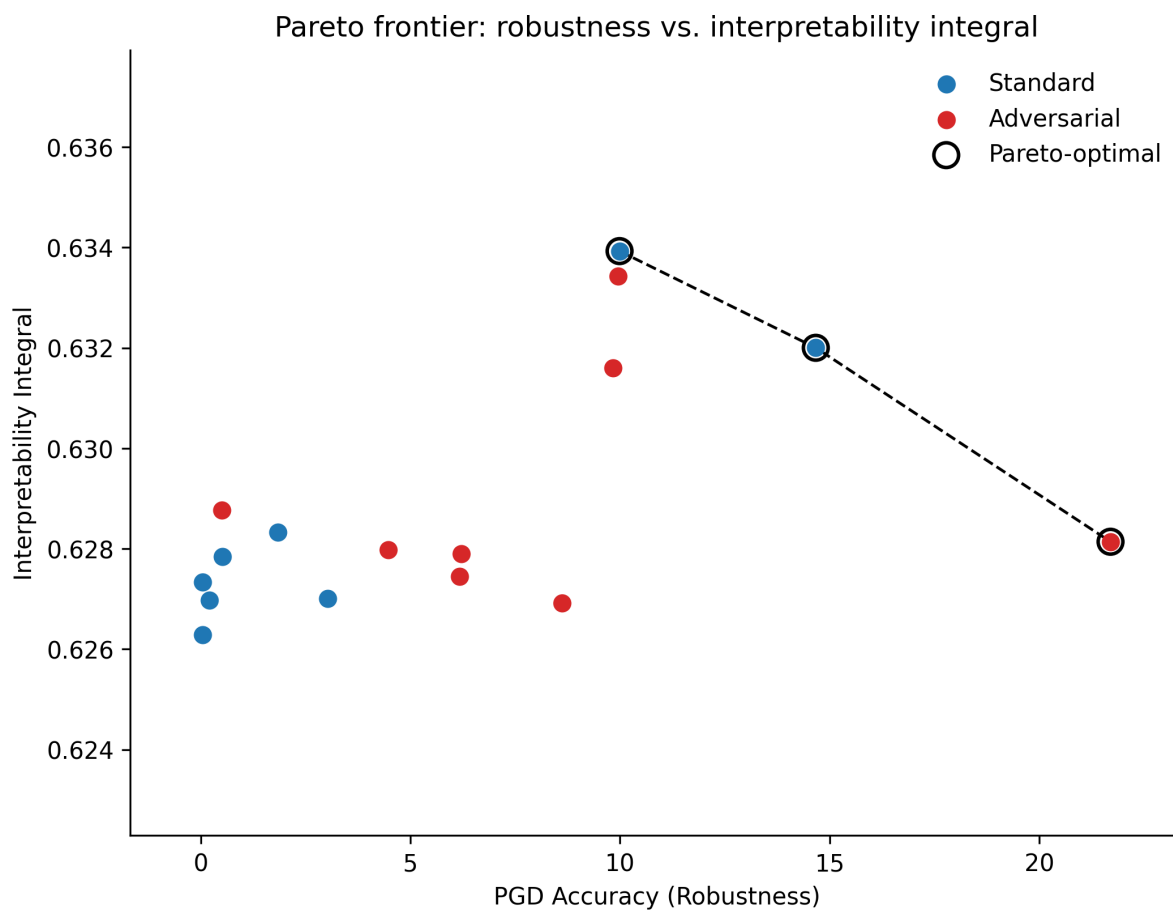


4.11 Pareto Frontier

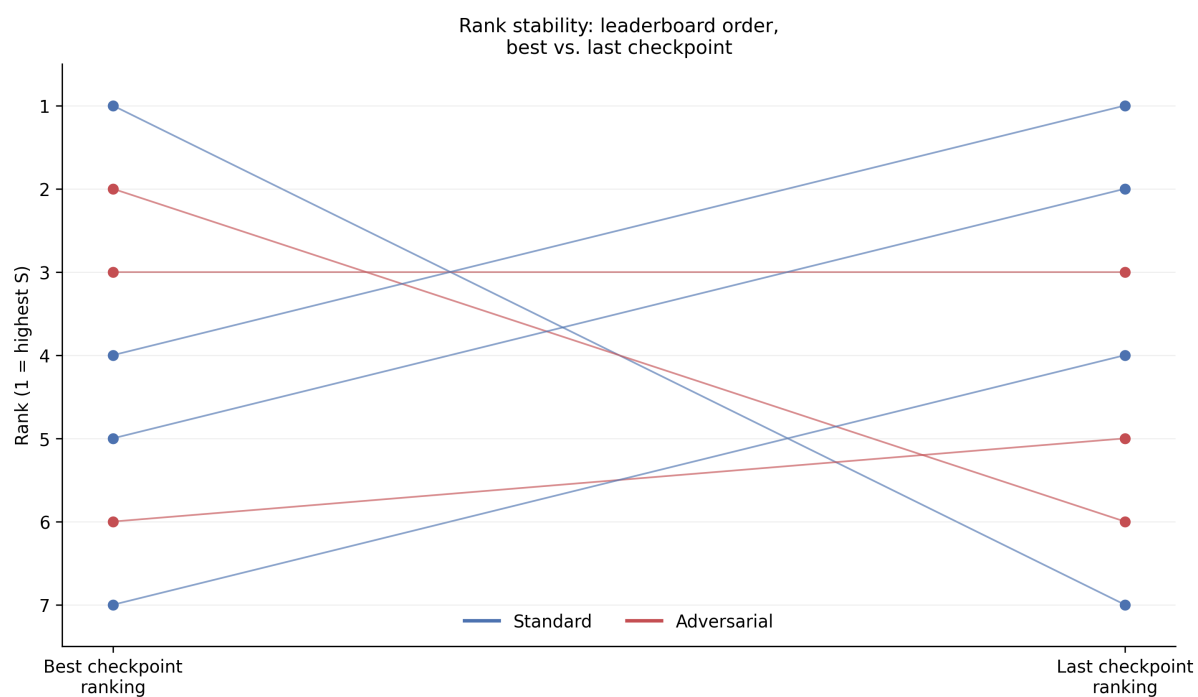
Figure 19 (10.3). Pareto frontier: robustness vs. interpretability



4.12 Pareto Frontier (Integral)



4.13 Rank Stability



4.14 Rank Stability (Integral)

